

Run-to-Run Consistency of Business Recommendations from Web-Search-Enabled Large Language Models: An Exploratory Study

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ABSTRACT. Consumers increasingly ask conversational large language models (LLMs) which product or company to choose, yet little is documented about how stable those recommendations are under repetition. We conducted an exploratory study of run-to-run consistency for two widely used web-search-enabled assistants, OpenAI's ChatGPT and Google's Gemini. Ten open "best X for a small business" questions spanning ten commercial categories were each issued three times to each assistant within a single day (60 responses total), with web search enabled throughout. From each response we extracted the set of recommended businesses and measured consistency as the mean pairwise Jaccard overlap of these sets across the three repetitions. Overall consistency was 69.5%, i.e. roughly one recommended business in three changed between identical queries. Consistency differed by assistant (ChatGPT $87.2\% \pm 17.5$; Gemini $51.9\% \pm 11.8$) but the instability was not uniform: of 127 distinct business-slots, 56% appeared in every repetition (a stable core) while 27% appeared in only one of three (a volatile tail). We discuss implications for reproducibility research and for the emerging practice of generative engine optimization, and we release the full method and response data. Given the small sample the results are directional, not definitive.

1. Introduction

Conversational assistants built on large language models are increasingly used as recommenders: a user asks for the "best" tool, service, or vendor in a category and receives a short list of named businesses. When these assistants are connected to live web search, the recommendation is assembled at query time from retrieved sources rather than recalled from static training data. A natural and practically important question follows: if the same user asks the same question more than once, how much does the recommended set change?

Stability matters for two audiences. For researchers studying LLM reliability, run-to-run variance under a fixed prompt is a form of reproducibility measurement. For businesses and the practitioners who advise them, the degree to which an assistant reliably names a given company determines whether "being recommended by AI" is a durable position or a transient one. This study offers an initial, deliberately small measurement of that variance, and characterizes its structure.

2. Method

2.1 Questions

We selected ten open-ended "best/top X for a small business" questions, one per commercial category: press-release distribution, customer relationship management, email marketing, project management, accounting, marketing agencies, website builders, live chat, e-commerce platforms, and appointment scheduling. Open questions were used deliberately; head-to-head comparison prompts (of the form "X versus Y") were excluded because they can only return the two names supplied in the prompt and therefore trivially maximize apparent consistency.

2.2 Assistants and procedure

Two assistants were queried: ChatGPT via the OpenAI Responses API with the web_search tool enabled, and Google Gemini via the Generative Language API with Google Search grounding enabled. All queries specified a United States locale and English-language responses. Each question was issued three independent times to each assistant, yielding 60 responses. All responses were collected on September 13, 2026 within a single session to hold model versions constant. On this run, all 60 responses were grounded in a live web search.

2.3 Extraction and measure

From each response we extracted the set of specific businesses recommended, using an automated language-model extraction pass instructed to include only named businesses presented as options and to exclude media outlets, generic categories, and services named only as context. We validated the extraction against the source text: all 290 extracted business names appeared verbatim in the response they were drawn from (no fabricated names), and manual inspection of a sample confirmed that section headings matched the extracted set.

For each question and assistant we computed consistency as the mean pairwise Jaccard similarity of the recommended sets across the three repetitions, counting only repetitions that produced a shortlist of two or more businesses (this prevents an empty or single-item response from being scored as a "different" recommendation). We separately counted, per question-assistant, how many distinct businesses appeared in all repetitions (the stable core) versus exactly one repetition (the volatile tail).

3. Results

Across all questions and both assistants, mean run-to-run consistency was **69.5%**. Equivalently, about one recommended business in three (approximately 31%) changed when the identical question was reissued minutes later. Consistency differed markedly between the two assistants, though with only ten questions per assistant these means carry no formal confidence interval and should be read as directional.

Assistant	Mean consistency	SD	Range	Shortlist rate
ChatGPT (web search)	87.2%	17.5	59-100%	100%
Gemini (Search grounding)	51.9%	11.8	33-67%	97%

Table 1. Run-to-run consistency by assistant (n = 10 questions each). Shortlist rate is the share of the 30 responses that named two or more businesses.

The instability was structured rather than uniform. Of 127 distinct business-slots observed across all question-assistant cells, **71 (56%)** were named in every repetition, and **34 (27%)** were named in only one of the three. Qualitatively, the leading one to three recommendations for a given question tended to recur across all repetitions, while lower-ranked positions rotated. The businesses named most often, after excluding any name appearing in a prompt, were Shopify (12 mentions), Wix (8), and Squarespace (7), followed by a group of twelve tied at six mentions each.

Question (category)	ChatGPT	Gemini
PR distribution	59%	50%
CRM	100%	42%
Email marketing	62%	63%
Project management	100%	67%
Accounting	100%	33%
Marketing agencies	89%	61%
Website builders	100%	62%
Live chat	100%	58%
E-commerce	100%	33%*
Scheduling	62%	50%

Table 2. Per-question consistency. With three repetitions the per-cell values fall on a coarse scale and are indicative only.

*The Gemini e-commerce cell rests on two repetitions; the third named a single business and was excluded by the two-or-more rule.

4. Discussion

Two observations stand out. First, the variance is real but bounded and structured: a stable core of leading recommendations recurs while a tail rotates. For the reproducibility literature this suggests that summarizing LLM recommendation stability with a single scalar understates the picture; the head and tail of a ranked recommendation behave differently. Second, the two assistants differ substantially in how much they permit the tail to move, which implies that consistency is a property of the assistant and its retrieval-and-composition pipeline, not only of the question.

For the practice of generative engine optimization, the structural finding has a direct reading: the positions a smaller or newer entrant can realistically reach (the lower-ranked slots) are precisely the least stable ones, whereas established names occupy the stable core. This is consistent with, though not proof of, the hypothesis that assistants converge on entities that are described consistently and frequently across independent sources.

5. Limitations

This is a small exploratory study. Ten questions, three repetitions, two assistants, and a single collection day are sufficient to surface a pattern but not to estimate precise category-level or assistant-level rates; the reported means carry no formal margin of error. With three repetitions, per-question consistency values are coarsely quantized. The automated extraction, while validated against the source text for fabricated names, can under- or over-count borderline mentions (for example treating "Wave" and "Wave Accounting" as distinct), which would tend to understate rather than inflate consistency. Findings reflect two specific assistants on one day; assistant behavior evolves over time. The study was conducted by a commercial entity with an interest in the subject matter; the method and data are released to permit independent verification.

6. Data and reproducibility

The ten questions, the three-repetition procedure, and the Jaccard-overlap measure specified above are sufficient to reproduce the study. The full set of 60 responses, the analysis code, and the methodology are openly available (CC BY 4.0) at github.com/jjoseph18/ai-recommendation-consistency, and the study is summarized at the companion page pressfront.co/ai-visibility-index.html.

References

1. OpenAI. *Responses API and the web_search tool*. API documentation, 2026.
2. Google. *Gemini API: grounding with Google Search*. Generative Language API documentation, 2026.
3. Jaccard, P. The distribution of the flora in the alpine zone. *New Phytologist*, 11(2):37-50, 1912.
4. Aggarwal, P., et al. Geo: Generative engine optimization. In *Proceedings of the ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2024.

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